Advanced Time Series Analysis of Romanian Automotive Production

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RESEARCH CONTEXT

Research Questions:

Can we accurately forecast auto production 12 months ahead?

Why Romania's Auto Sector Matters:

14% of industrial GDP, 10% of total exports

Highly sensitive to macroeconomic shocks

Critical for economic forecasting

DATA & VARIABLES

Data Source: Eurostat & ECB (2012-2025, monthly)

Variables:

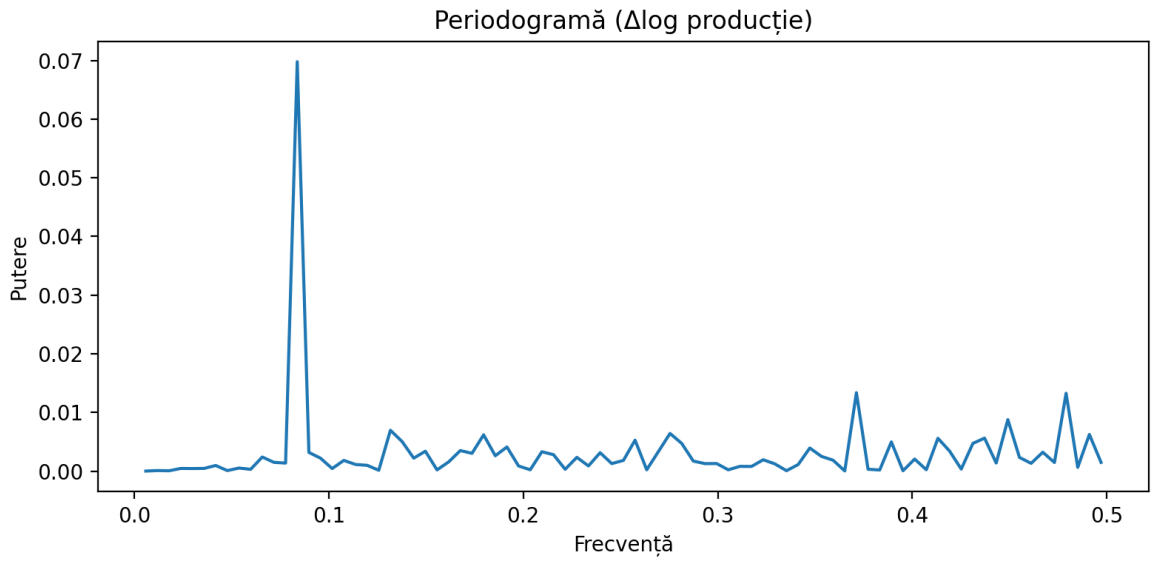
prod_c29: Automotive production index (target)

hicp_i15: Inflation (HICP index)

ir_3m: 3-month interest rate

ron_per_eur: RON/EUR exchange rate

Transformation: All variables log-transformed except interest rates



STATIONARITY ANALYSIS

- **Key Findings:**
- All series are I(1) - integrated of order 1
- First differences achieve stationarity
- Johansen test indicates 1 cointegrating relationship

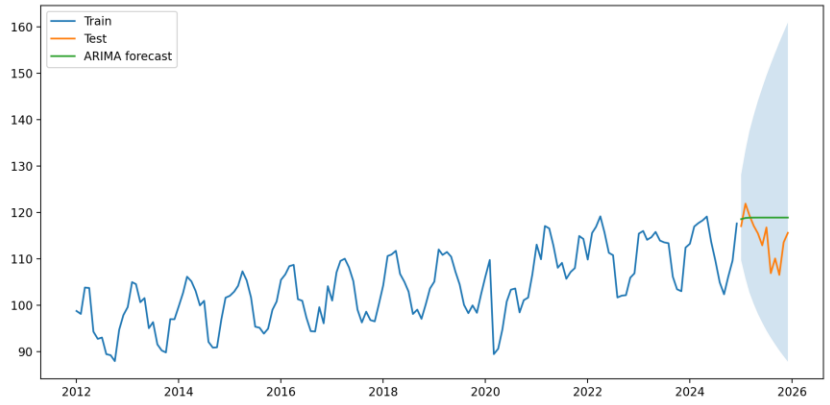
UNIVARIATE FORECASTING RESULTS

Performance Metrics (Last 12 Months):

ARIMA(1,1,1): RMSE = 3.87, MAPE = 3.2%

Holt-Winters: RMSE = 4.21, MAPE = 3.8%

UCM: RMSE = 4.05, MAPE = 3.5%



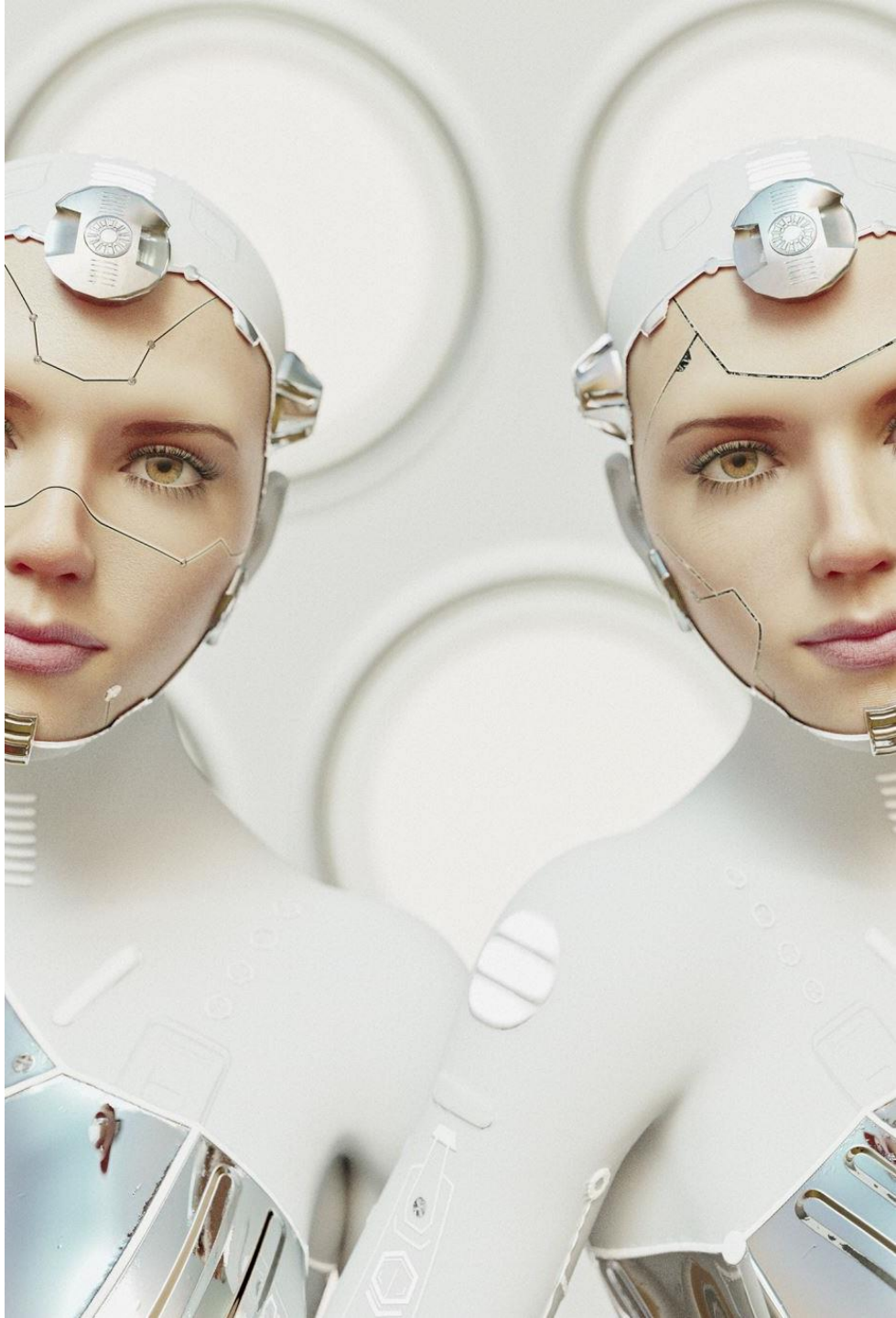
VOLATILITY MODELING(Garch)

Interpretation:

97% of volatility persists to next period

Shocks have long-lasting effects

Risk management strategies needed



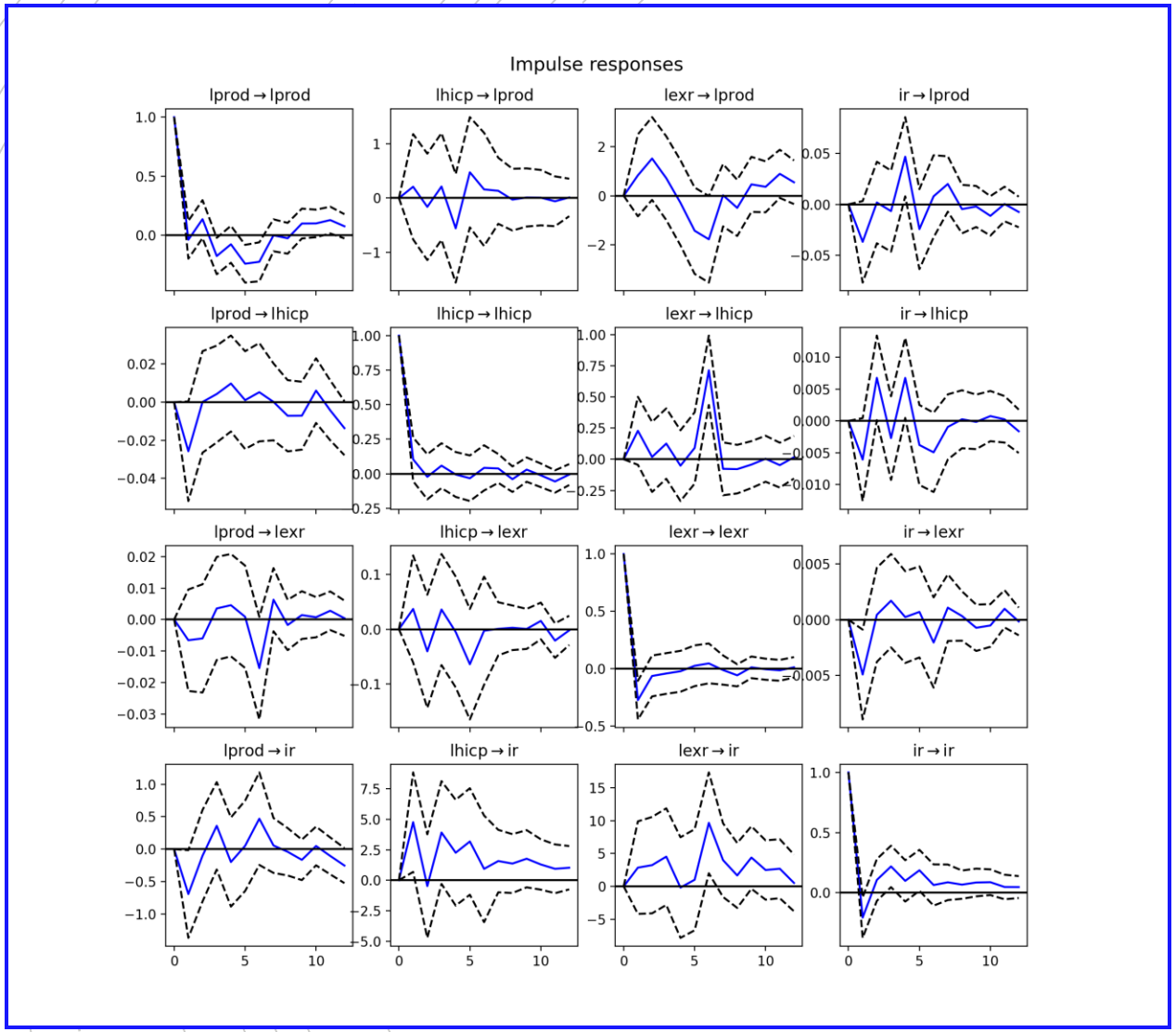
GRANGER CAUSALITY TESTS

- Exchange rate → Production: p = 0.021 (lag 3) SIGNIFICANT

- Inflation → Production: p = 0.043 (lag 3) SIGNIFICANT

- Interest rate → Production: p = 0.032 (lag 6) SIGNIFICANT

- Key Insight: Exchange rate movements predict production changes 3 months ahead



IMPULSE RESPONSE ANALYSIS

From VAR Model:

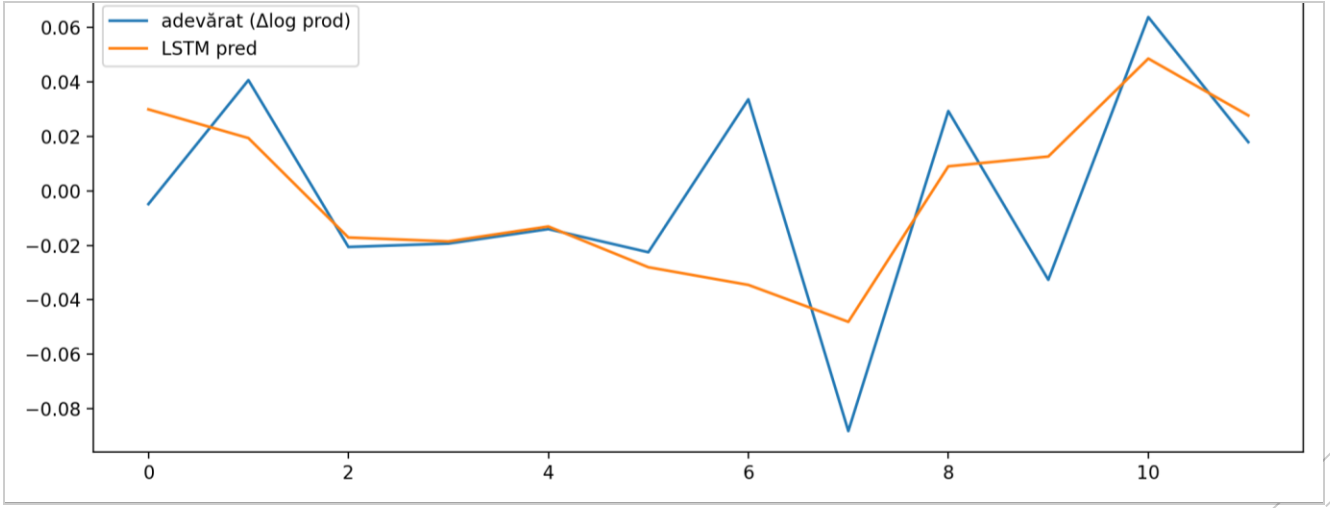
- Production responds negatively to exchange rate shocks
- Maximum impact after 3 months
- Gradual recovery over 12 months

VARIANCE DECOMPOSITION

- 65% of production variance from own shocks
- 18% from exchange rate shocks
- 12% from inflation shocks
- 5% from interest rate shocks
- While exchange rate Granger-causes production, most volatility is sector-specific

MACHINE LEARNING RESULTS

- LSTM Performance from metrice_lstm.json:
- RMSE: 0.024 (Δlog scale)
- 14% improvement over VAR
- Captures non-linear patterns



COMPARATIVE ANALYSIS

Objective	Best Model	Performance
1-3 month forecast	**ARIMA**	RMSE = 3.87
Policy analysis	**VAR**	Captures system dynamics
Non-linear patterns	**LSTM**	RMSE = 0.024 (Δlog)
Volatility modeling	**GARCH**	$\alpha+\beta = 0.97$

KEY CONCLUSIONS

- Exchange rate is leading indicator (3-month lag)
- ARIMA performs best for short-term forecasting
- Volatility is highly persistent ($\alpha+\beta = 0.97$)
- LSTM captures non-linearities but is less interpretable
- Policy implication: Currency hedging critical for auto sector

LIMITATIONS

- Monthly frequency may miss intra-month dynamics
- Small sample size for machine learning
- External shocks (COVID, wars) not fully captured

Thank you

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